1. In active voice, the grammatical subject of the sentence:

2. In passive voice, the grammatical subject of the sentence:

3. Identify the sentence written in passive voice:

4. Which auxiliary verb is fundamental to forming all passive voice constructions?

5. What is the passive form of: 'The enzyme cleaves the substrate'?

6. What is the passive form of: 'The technician sterilized the glassware'?

7. An agentless (truncated) passive sentence is one where:

8. Why is the agentless passive dominant in Materials and Methods sections?

9. Which modal auxiliary expresses absolute necessity or strict regulatory command?

10. Which modal auxiliary expresses tentative possibility in scientific claims?

11. In 'Bacterial cells can synthesize essential amino acids', 'can' expresses:

12. Which modal is traditionally used to express formal requirement in patents and standards?

13. Choose the correct passive: 'Scientists are developing a novel biosensor.'

14. Choose the correct passive: 'The team has synthesized three peptide analogs.'

15. Choose the correct modal for logical deduction: 'The culture is turbid; bacteria ___ be growing.'

16. Which modal indicates advice or standard operational recommendation?

17. In 'You may now remove the tubes from the thermal cycler', 'may' expresses:

18. What is the passive form of: 'Researchers will evaluate the pharmacokinetic profile'?

19. Which sentence is an impersonal passive construction?

20. Which modal expresses past unfulfilled possibility?

21. Choose the correct modal: 'The sample ___ have been contaminated during vortexing.'

22. Can intransitive verbs (e.g., deteriorate, arrive, sleep) be converted into passive voice?

23. Why cannot intransitive verbs be converted into passive voice?

24. What is the passive form of: 'The centrifuge had spun the tubes for 10 minutes'?

25. Which modal auxiliary denotes past habit or routine action in research?

26. In 'The mutation could disrupt the active site', 'could' functions as an:

27. Which semi-modal is synonymous with 'must' in expressing external obligation?

28. Choose the correct negative modal for prohibition in biosafety:

29. In 'The results should be reproducible under standard conditions', 'should' expresses:

30. What is the passive form of modal verbs: 'The team must autoclave all biohazardous waste'?

31. Identify the modal of deduction indicating high improbability:

32. In 'This band cannot be the target amplicon because its size is 1,000 bp', 'cannot' expresses:

33. What is the passive form of: 'The enzyme can hydrolyze cellulose'?

34. Which of the following sentences uses active voice effectively to highlight agency?

35. In 'Cells ought to be incubated immediately', 'ought to' conveys:

36. Choose the correct passive: 'Someone left the incubator door open.'

37. Which modal replaces 'can' in past tense to denote general past ability?

38. Choose the correct phrase: 'The researchers were ___ isolate the intact plasmid.'

39. In 'The protocol might require optimization', 'might' indicates:

40. Identify the sentence that contains both a modal auxiliary and passive voice:

41. Select the sentence where passive voice is rhetorically preferred over active voice:

42. Which sentence demonstrates proper scientific hedging using modal verbs?

43. What is the passive voice of: 'The Principal Investigator gave the postdoc a new plasmid'?

44. Why can ditransitive verbs (give, assign, grant) generate two passive forms?

45. Choose the correct modal: 'The frozen cells ___ thawed rapidly at 37 deg C to preserve viability.'

46. Identify the sentence using an impersonal passive reporting structure:

47. Choose the correct passive: 'Nobody has ever observed this anomalous phenotype.'

48. In 'The protein would precipitate if salt were added', 'would' is used as:

49. Choose the correct past modal: 'We didn't detect the band; the probe ___ degraded.'

50. In 'The probe must have degraded', 'must have' indicates:

51. Choose the correct past modal: 'You ___ checked the pH before adding the enzyme; it was too acidic!'

52. In 'You should have checked the pH', 'should have' expresses:

53. Select the sentence with an ergative verb used intransitively without passive marking:

54. Choose the correct passive: 'They are to publish the clinical trial protocol next month.'

55. Identify the sentence that improperly uses passive voice, producing unnecessary wordiness:

56. Choose the correct modal: 'All biosafety cabinets ___ undergo annual HEPA recertification.'

57. In 'Preliminary data indicate that compound X could potentially inhibit viral fusion', what is stylistically flawed?

58. Select the sentence where active voice is preferred to emphasize authorial responsibility:

59. What is the passive form of: 'The lab must have discarded the expired reagents'?

60. Choose the correct modal: 'You ___ not touch the acrylamide gel without nitrile gloves!'

61. Identify the sentence where 'can' expresses epistemic possibility rather than dynamic ability:

62. Choose the correct passive: 'Researchers have to monitor bioreactor temperature constantly.'

63. In 'The enzyme is thought to interact with membrane lipids', what passive construction is used?

64. Transform into active voice: 'A significant increase in absorbance was observed by the team.'

65. Choose the correct modal: 'If the primer is degraded, the PCR ___ fail.'

66. Select the sentence with a pseudo-passive (prepositional passive):

67. Why is 'The protocol was agreed upon' grammatically valid in English?

68. Choose the correct modal: 'Cryo-EM ___ resolve macromolecular structures at near-atomic resolution.'

69. Identify the sentence that correctly uses passive voice with a causative verb:

70. In active voice, 'make' takes a bare infinitive ('We made the cells express'); in passive, it takes:
'The cells were made to express') (B) A bare infinitive

71. Choose the correct modal: 'The autoclave steam valve ___ be inspected weekly per OSHA standards.'

72. Identify the sentence containing an erroneous double modal:

73. Why is 'might could' ungrammatical in standard English syntax?

74. Choose the correct passive: 'One can deduce the molecular weight from migration distance.'

75. Select the sentence that avoids misplaced agency in passive construction:

76. Identify the sentence where passive voice introduces dangerous syntactic ambiguity:

77. Select the sentence that deploys epistemic modality with maximal academic sophistication:

78. Choose the correct modal for a hypothetical retrospective concession: 'Even if the probe had bound, the signal ___ have been undetectable due to quenching.'

79. What is the passive form of: 'The Principal Investigator made the graduate student re-run the electrophoresis gel'?

80. Select the sentence where active voice is essential to prevent a dangling agent:

81. In 'It is recommended that the reaction mixture ___ incubated on ice':

82. Choose the correct modal expression: 'The laboratory ___ decontaminate all surfaces before exiting BSL-3.'

83. Identify the sentence containing an erroneous passive transformation of a stative verb:

84. Why cannot 'The flask was contained 500 mL' be accepted?

85. Choose the correct past modal: 'The wild-type colonies grew, so the ampicillin plate ___ lacked the antibiotic.'

86. Select the sentence that effectively mitigates over-claiming (hedging) in Discussion sections:

87. What is the passive form of: 'They let the culture incubate for 48 hours'?

88. Why is 'The culture was let incubate' avoided in formal English?

89. Choose the correct modal: 'The bio-sensor ___ detect antigens at concentrations as low as 1 pg/mL.'

90. Select the sentence with a dangling passive infinitive:

91. Choose the correct modal: 'Judging from the sharp melting peak, the amplicon ___ be highly specific.'

92. Identify the sentence that correctly transforms an indirect object into a passive subject:

93. Choose the correct modal: 'The researcher ___ have calibrated the balance; all weighings are flawed!'

94. In 'The protein was found to be phosphorylated at Ser-128', what is the active equivalent?

95. Select the sentence that avoids an ungrammatical 'by'-phrase with an inanimate instrument:

96. Why is 'with a scalpel' grammatically correct while 'by a scalpel' is flawed?

97. Choose the correct modal: 'Had the enzyme been folded, it ___ have exhibited catalytic turnover.'

98. Identify the sentence demonstrating flawless integration of modal hedging and passive voice:

99. Choose the correct modal: 'The bio-safety cabinet airflow ___ not be obstructed under any operational circumstances.'

100. Identify the sentence that achieves impeccable syntactic balance between active authorial agency and passive methodological objectivity:

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Analyze the syntactic mechanics of voice transformation from active to passive across the full spectrum of English tenses.

Q2. Examine the rhetorical, scientific, and cognitive justifications for the dominance of the agentless passive in experimental methodologies.

Q3. Discuss the legitimate role of the active voice in contemporary scientific papers, analyzing guidelines from Nature, Science, and NIH.

Q4. Analyze impersonal passive constructions ('It is reported that...', 'It was observed that...') and evaluate their stylistic impact.

Q5. Examine the taxonomy of English modal auxiliaries, distinguishing epistemic, deontic, and dynamic modalities in science.

Q6. Analyze modal hedging techniques in scientific Results and Discussion sections to calibrate empirical claims.

Q7. Detail the syntax and semantics of past modal constructions (Could have, Might have, Must have, Should have) in experimental troubleshooting.

Q8. Examine the syntax of ditransitive verbs in passive voice, analyzing direct versus indirect object passivization.

Q9. Analyze prepositional passives (pseudo-passives) and the phenomenon of preposition stranding in technical prose.

Q10. Examine ergative (middle) verbs versus passive voice in describing biochemical and biophysical state changes.

Q11. Discuss the Mandative Subjunctive versus modal periphrastics ('should') in international biosafety standards.

Q12. Analyze the misuse of 'can' versus 'may' in scientific laboratory permissions and protocol documentation.

Q13. Examine passive transformations of causative verbs ('make', 'let', 'have') in experimental methodology descriptions.

Q14. Analyze the syntax and semantics of 'Must not' versus 'Do not have to' in biosafety containment levels.

Q15. Examine the dangling agent problem in passive constructions with introductory purpose or participial phrases.

Q16. Analyze the epistemic strength of modal verbs in constructing formal scientific hypotheses.

Q17. Detail the passive voice of phrasal verbs, prepositional verbs, and multi-word verbs in biotechnology literature.

Q18. Examine the passive voice with modal perfects (Must have been, Could have been, Should have been) in retrospective audit reports.

Q19. Discuss the stylistic peril of 'Passive Stacks' (chained passive clauses) and demonstrate sentence disentangling.

Q20. Analyze modal verbs of deduction in interpreting negative controls and experimental anomalies.

Q21. Examine the syntax of 'Need' and 'Dare' as marginal modal auxiliaries versus lexical main verbs.

Q22. Analyze the passive voice in describing flowcharts, manufacturing pipelines, and bioprocess engineering flow.

Q23. Examine the modal 'Would' in describing past laboratory routines, prospective designs, and polite discourse.

Q24. Analyze the passive transformation of verbs taking that-clause complements and infinitive raising.

Q25. Examine the modal auxiliary 'Shall' in legal patents, regulatory compliance, and scientific declarations.

Q26. Analyze the interaction between negation and modal verbs (Epistemic vs. Deontic Scope of Negation).

Q27. Discuss the passive voice in writing Material Transfer Agreements (MTAs) and Non-Disclosure Agreements (NDAs).

Q28. Examine the avoidance of passive voice with reciprocal verbs and non-passivizable idioms in technical discourse.

Q29. Analyze the modal verb 'Used to' versus 'Would' in historical accounts of biotechnological breakthroughs.

Q30. Synthesize guidelines for achieving stylistic harmony between active voice, passive voice, and modal hedging in a doctoral dissertation.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1: [A]
Q2: [B]
Q3: [B]
Q4: [B]
Q5: [A]
Q6: [B]
Q7: [B]
Q8: [B]
Q9: [C]
Q10: [B]
Q11: [A]
Q12: [A]
Q13: [B]
Q14: [B]
Q15: [A]
Q16: [A]
Q17: [A]
Q18: [B]
Q19: [A]
Q20: [A]

Q21: [A]
Q22: [B]
Q23: [A]
Q24: [A]
Q25: [A]
Q26: [A]
Q27: [A]
Q28: [B]
Q29: [A]
Q30: [B]
Q31: [A]
Q32: [A]
Q33: [A]
Q34: [A]
Q35: [A]
Q36: [A]
Q37: [A]
Q38: [A]
Q39: [A]
Q40: [A]

Q41: [B]
Q42: [B]
Q43: [C]
Q44: [A]
Q45: [A]
Q46: [A]
Q47: [A]
Q48: [A]
Q49: [A]
Q50: [A]
Q51: [A]
Q52: [A]
Q53: [D]
Q54: [A]
Q55: [A]
Q56: [A]
Q57: [A]
Q58: [A]
Q59: [A]
Q60: [A]

Q61: [A]
Q62: [A]
Q63: [A]
Q64: [A]
Q65: [A]
Q66: [A]
Q67: [A]
Q68: [A]
Q69: [A]
Q70: [A]
Q71: [A]
Q72: [A]
Q73: [A]
Q74: [A]
Q75: [B]
Q76: [A]
Q77: [A]
Q78: [A]
Q79: [A]
Q80: [B]

Q81: [A]
Q82: [A]
Q83: [A]
Q84: [A]
Q85: [A]
Q86: [B]
Q87: [A]
Q88: [A]
Q89: [A]
Q90: [A]
Q91: [A]
Q92: [C]
Q93: [A]
Q94: [A]
Q95: [B]
Q96: [A]
Q97: [A]
Q98: [A]
Q99: [A]
Q100: [A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

[illegible]

- Preserving Authorial Anonymity in Blind Peer Review:
- Agentless passives prevent accidental revelation of laboratory identities in double-blind submission processes.

[illegible]

- Overuse of impersonal passives clutters text with empty introductory phrases:
- Bloated: "It was observed by the investigators that the cells underwent apoptosis."
- Streamlined Active: "The cells underwent apoptosis."
- Streamlined Methodological: "Apoptosis was observed in 80% of treated cells."

[illegible] in science.

Combining multiple hedging devices in the same clause produces weak, unscholarly prose:
- Flawed: "These data might possibly perhaps indicate a potential binding interaction."
- Revised: "These data suggest that a binding interaction may occur."

[illegible]

thematic topic of the preceding discourse.

[illegible]

[illegible]

In active voice, 'make', 'let', and 'have' govern an object followed by a BARE INFINITIVE (without 'to'):
- "We made the cells express GFP."
- "The technician let the agar solidify."
- "The PI had the postdoc sequence the clones."

2. Morphosyntactic Realignment in Passive Voice:
- Passive of 'Make' (Requires restoring 'to'-infinitive):
- Active: "The osmotic shock made the bacterial cells burst."
- Passive: "The bacterial cells were made to burst by the osmotic shock." (Mandatory 'to').
- Passive of 'Let' (Suppletive replacement with 'allow to'):
- 'Let' cannot be passivized directly (*The agar was let solidify is ungrammatical).
- Passive: "The agar was allowed to solidify for 30 minutes."
- Passive of 'Have' (Suppletive replacement with 'instruct to' or 'ask to'):
- Passive: "The postdoc was instructed to sequence the clones."

[illegible]

3. Three Remediation Strategies:
- Strategy 1 (Active Voice with Human Agent):
- "To assess microbial viability, the researchers incubated the agar plates for 24 hours."
- Strategy 2 (Recasting Introductory Phrase with Explicit Passive):
- "For microbial viability to be assessed, the agar plates were incubated for 24 hours."
- Strategy 3 (Nominalization of Purpose):
- "Assessment of microbial viability was achieved by incubating the agar plates for 24 hours."

>> Solution to Question [illegible] [illegible] literature.

[illegible] audit reports.

2. Modal Formulations in Control Interpretation:
- Deduction of Contamination (Positive Signal in Negative Control):
- "Because the non-template control lane produced a bright 500 bp band, the PCR master mix must be contaminated with exogenous plasmid template." (High certainty; 'must be').
- Deduction of Reagent Failure (Negative Signal in Positive Control):
- "Because the positive control failed to fluoresce, the primary antibody must have lost its binding activity."
- Deduction of Impossibility (Cannot / Can't):
- "The observed 25 kDa band cannot be the full-length recombinant fusion protein, which has a calculated molecular weight of 72 kDa." (Logical impossibility).
- Tentative Inference (May / Might):
- "The faint band at 25 kDa may represent a proteolytic degradation fragment or an alternative splice variant."

[illegible]

vessel."
- "Glucose feed and dissolved oxygen levels are maintained via automated feedback loops."
- "Following stationary phase arrest, the broth is clarified by continuous centrifugation."
- "Target antibodies are subsequently captured using Protein A affinity chromatography, after which viral inactivation is performed at low pH."

[illegible]

In contemporary spoken and academic English, 'shall' has virtually disappeared as a first-person future marker, replaced universally by 'will'.

2. The Specialized Bastion of 'Shall': Legal Contracts & Patent Claims:
In regulatory frameworks (FDA, EMA), international treaties, and patent claims, 'shall' carries immense, legally binding deontic force:
- Mandatory Patent Requirement: "The lyophilized formulation shall contain between 5% and 10% trehalose by weight."
- Legal Meaning: A formulation with 4.9% trehalose falls outside patent coverage; 'shall' establishes strict legal boundary conditions.
- Statutory Regulatory Compliance: "The sponsor shall submit annual safety reports to the regulatory authority."
- Meaning: Absolute legal obligation; failure constitutes a statutory offense.

[illegible]

- "Upon termination of the research project, all remaining aliquots of the Material shall be autoclaved and destroyed, or returned to the Provider upon request."
- Confidentiality and Intellectual Property:
- "Confidential Information shall be maintained under strict secrecy, and shall be disclosed only to employees with a verified need to know."
- Function: Establishes impersonal, institutional obligations binding upon corporations rather than individual transient researchers.

[illegible] discourse.

[Problem Statement]: Synthesize guidelines for achieving stylistic harmony between active voice, passive voice, and modal hedging in a doctoral

[Detailed Step-by-Step Answer]:

1. The Rhetorical Symphony of a Dissertation:

A doctoral thesis in biotechnology is an advanced scientific argument requiring a calibrated blend of active agency, objective passivity, and epistemic modesty:

2. Section-by-Section Best Practices:

- Abstract & Title: Active and direct. State the core finding immediately: "Here, we show that..."
- Introduction: Active voice when positioning your doctoral hypothesis against literature; passive when summarizing broad established scientific paradigms.
- Materials and Methods: 85-90% agentless passive voice to ensure total methodological objectivity and universal reproducibility; active voice only when detailing novel customized protocols.
- Results: Balanced alternation. Passive for procedural setups ("Cells were treated with compound 4b"); Active for observational outcomes ("The compound elicited a 3-fold reduction in viability; Figure 3 illustrates this trend").
- Discussion & Conclusions: Modals and hedging become paramount (may, could, should, suggest, indicate) to interpret biological implications with scholarly rigor, avoiding dogmatic over-claiming while actively defending authorial interpretations.
